

VERIFYX

POINT. VERIFY. FIX. VERIFY AGAIN.

AI-Powered Physical-World Verification Engine

PHONE-FIRST • OFFLINE-FIRST • LOCAL AI VISION

THE PROBLEM

THE STANDARD IS DIGITAL. THE WORKPLACE IS PHYSICAL.

01. CHECKLIST

02. WALK

03. OBSERVE

04. PHOTOGRAPH

05. COMPARE

06. REPORT

MANUAL

Requires human inspectors to physically check every location manually.

SLOW

Compliance reports take hours or days to compile and submit.

INCONSISTENT

Subjective human checks lead to missed hazards and variance.

SCATTERED EVIDENCE

Inspection photos are isolated without structured audit context.

"Businesses have standards. People still have to manually verify reality."

THE SOLUTION

TURN THE PHONE INTO A VERIFICATION TOOL.

STEP 01

POINT

Aim camera or select evaluation scenario in workplace zone.

STEP 02

VERIFY

Local AI evaluates physical evidence against inspection rules.

STEP 03

FIX

Interactive visual feedback guides user to clear hazards.

STEP 04

VERIFY AGAIN

Re-scan confirms clearance & locks immutable evidence.

VERIFYX observes the physical environment, evaluates evidence against inspection rules, and guides the user toward a verifiable result.

HOW IT WORKS

PERCEPTION → REASONING → EVIDENCE

01. CAMERA / INPUT

Live feed or approved test sample

02. YOLO11n (ONNX)

10.15 MB WASM local model

03. OBJECT DETECTION

Raw bounding box & class label

04. CONFIDENCE POLICY

Confidence tiers (HIGH/MED/LOW)

05. SPATIAL RULE ENGINE

Corridor overlap & geometry reasoning

06. VERIFICATION DECISION

PASS / REVIEW / RESCAN state

07. EVIDENCE HISTORY

Initial violation & re-scan frames

08. INSPECTION REPORT

Printable audit record & JSON output

LOCAL INFERENCE • ASYNC SINGLE-FLIGHT PROCESSING • NO MANDATORY CLOUD AI

DETECTION IS NOT VERIFICATION.

AI PERCEPTION (YOLO11n)

"What is visible?"

- Identifies presence of trained object classes
- Generates raw 2D bounding box coordinates
- Outputs raw model probability / confidence
- Single-frame static observation snapshot
- Lacks awareness of workplace compliance rules

VERIFYX SPATIAL RULE ENGINE

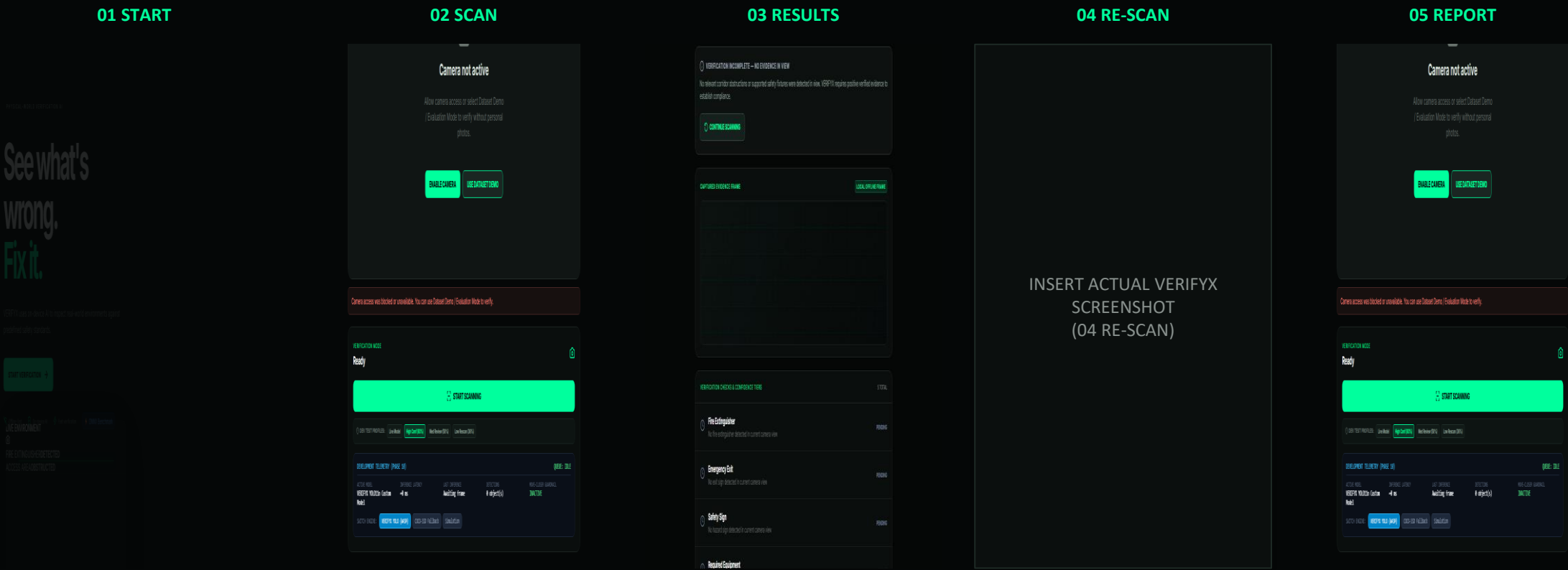
"Does the evidence satisfy the rule?"

- ✓ Enforces strict confidence tiers (HIGH/MED/LOW)
- ✓ Applies object-size guardrails (< 32px rescan)
- ✓ Evaluates 2D spatial pathway corridor overlap
- ✓ Requires multi-observation temporal confirmation
- ✓ Preserves baseline & re-scan history side-by-side

CORE GUARANTEE: INSUFFICIENT EVIDENCE NEVER BECOMES AN AUTOMATIC PASS.

WORKING PROTOTYPE

FROM SCAN TO VERIFIED INSPECTION REPORT



Dataset Demo / Evaluation Mode provides reproducible demonstration using approved held-out test imagery.

EXPERIMENTAL VALIDATION

MEASURED. TESTED. REPRODUCIBLE.

93.2%

FIRE EXTINGUISHER RECALL

Primary inspection zone (1.0m – 4.5m)

87.2%

EMERGENCY EXIT SIGN RECALL

Standard illuminated overhead signage

98.0%

HAZARD SIGN RECALL

Floor warning cones & safety placards

24 / 24

AUTOMATED TESTS PASSED

13 Spatial + 11 Report tests (100% pass)

10.15 MB

ONNX MODEL SIZE

640x640 resolution local WASM model

~2 FPS

BACKGROUND INFERENCE

~504ms latency with smooth 60 FPS video

** Measured evaluation results on prototype evaluation pool. Not certified compliance system.*

FROM PROTOTYPE TO PHYSICAL-WORLD INFRASTRUCTURE

NOW (PROTOTYPE)

- Local browser ONNX inference
- Workplace safety verification
- Spatial pathway corridor reasoning
- Offline print & JSON report lifecycle

NEXT HORIZONS

- Native Android deployment
- Snapdragon NPU acceleration
- Expanded inspection rule profiles
- iQOO Office Kit companion bridge

ENTERPRISE VISION

- Custom inspection template builder
- Multi-location audit compliance
- Broader physical-world verification
- Autonomous drone/robot vision integration

"VERIFYX bridges the gap between digital standards and the physical world."

VERIFYX • POINT. VERIFY. FIX. VERIFY AGAIN. • Prototype only — not a certified workplace safety compliance system.